

Guidewire Claim Intelligence - ML-Powered Claim Triaging

Maharshi Soni | machine-learning | advanced

Description

ML-powered claim intelligence engine for P&C insurance. Predicts claim severity, recommends adjuster assignment, estimates reserves, and flags fraud risk using anomaly detection. Trained on synthetic ClaimCenter data covering Auto, Homeowners, Workers Comp, and more.

Tech Stack

scikit-learn, pandas, numpy, fastapi, uvicorn, pydantic, matplotlib, pytest

Features

- Claim severity prediction (1-5 scale)
- Adjuster recommendation by expertise
- Reserve estimation
- Fraud risk scoring with Isolation Forest
- FastAPI REST API
- Feature importance explanations

Sample Input

Claim attributes

```
curl -X POST /score -d '{"lob": "PersonalAuto", "loss_amount": 15000}'
```

Sample Output

Claim intelligence scores

```
{"severity": 3, "fraud_score": 12, "adjuster": "Auto Specialist", "reserve": 18500}
```

Project Files

```
LICENSE
project-idea.json
pyproject.toml
README.md
requirements.txt
setup.py
.github/workflows/test.yml
src/claim_intelligence/config.py
src/claim_intelligence/data_generator.py
```

```
src/claim_intelligence/explanations.py
src/claim_intelligence/feature_engineering.py
src/claim_intelligence/scoring.py
src/claim_intelligence/__init__.py
src/claim_intelligence/api/app.py
src/claim_intelligence/api/schemas.py
src/claim_intelligence/api/__init__.py
src/claim_intelligence/models/adjuster_recommender.py
src/claim_intelligence/models/fraud_model.py
src/claim_intelligence/models/reserve_model.py
src/claim_intelligence/models/severity_model.py
src/claim_intelligence/models/__init__.py
tests/conftest.py
tests/test_adjuster_recommender.py
tests/test_api.py
tests/test_data_generator.py
tests/test_explanations.py
tests/test_feature_engineering.py
tests/test_models.py
tests/test_scoring.py
tests/__init__.py
```

Test Results

PASSED

Run: 61 | Passed: 61 | Failed: 0

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